Causal Inference Homework 1

Replication homework based on Cheng & Hoekstra (2012)

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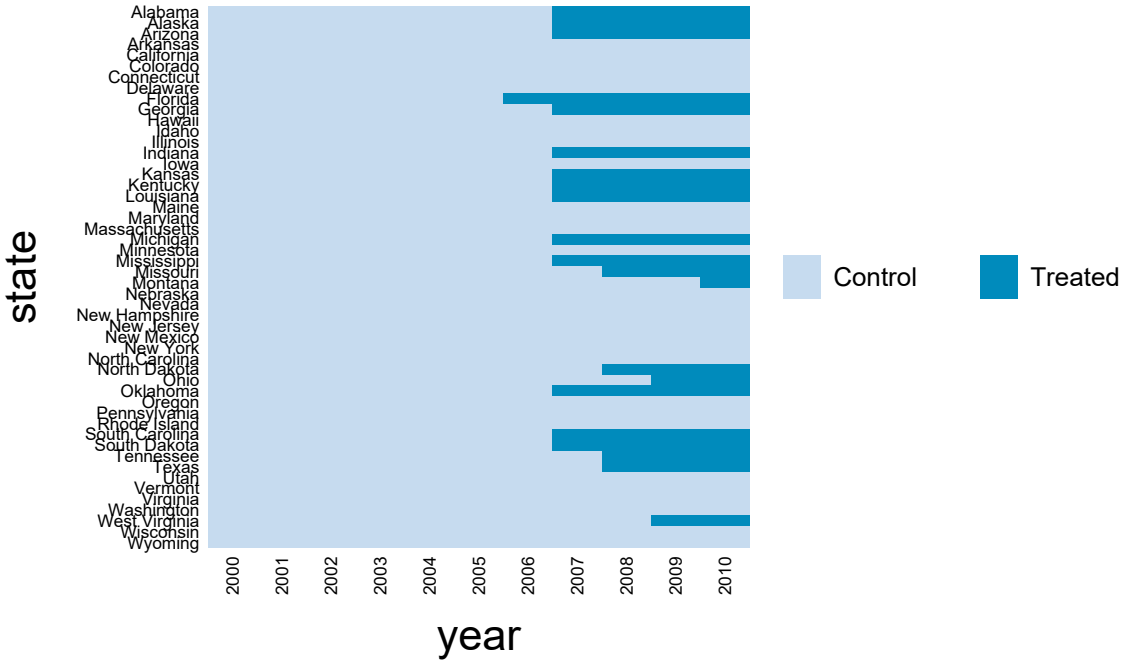
1 Data

Pull data from:

https://github.com/scunning1975/mixtape/raw/master/castle.dta

2 Graph of Treatment Status

As the table shows, most treatments occur in the year 2007. Because of this, we will mostly focus on the 2007 treatment in a non-staggered setting (through dropping states treated at other times).



3 DiD: Basic

To run a non-staggered causal study I only keep the states that are either treated in 2007 (first full year of treatment) or are never treated. I will not explicitly state this every time but this holds for all the DiDs and event studies until CSDiD at the end. Running simple reg and clustering standard errors at the unit (state) level over the two year window (pre: 2005, post: 2009) gives the following results:

assault	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
1.treatment_group	73.42783	45.29286	1.62	0.113	-18.04295	164.8986
1.after	-12.87794	5.347615	-2.41	0.021	-23.67767	-2.078211
treatment_group#after						
1 1	1.989454	10.78919	0.18	0.855	-19.79976	23.77867
_cons	236.7767	21.26314	11.14	0.000	193.8349	279.7185

Figure 1: Assault DiD Results

burglary	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
1.treatment_group	162.3379	71.96828	2.26	0.029	16.99498	307.6808
1.after	-42.15383	15.17633	-2.78	0.008	-72.80306	-11.50461
treatment_group#after						
1 1	64.42519	25.90661	2.49	0.017	12.10572	116.7447
_cons	653.6741	44.34778	14.74	0.000	564.112	743.2363

Figure 2: Burglary DiD Results

As the summary tables show, the effect of Castle Laws on assault are insignificant (p=0.855) but suggests an increase in assaults by a magnitude of around 2 per 100,000 people. The effect on burglary does appear significant (p=0.017) and suggests the laws are associated with an increase of 64 burglaries per 100,000 people in each state.

4 Event Study: Basic

To code in the leads and lags for the event study I added "pre_i" and "post_i" dummy variables (i representing period before/after treatment). I then ran a reghdfe and then a coefplot to visualize the results which gave the following event study plots:

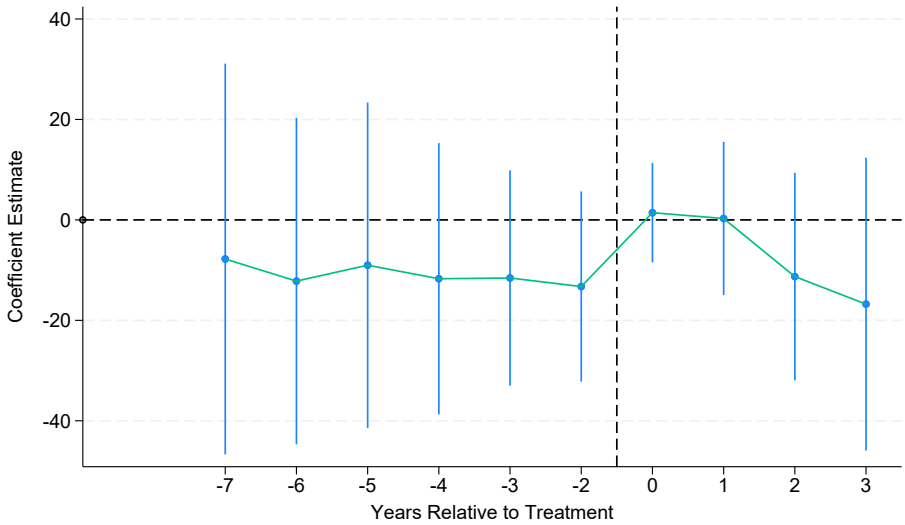


Figure 3: Assault Event Study

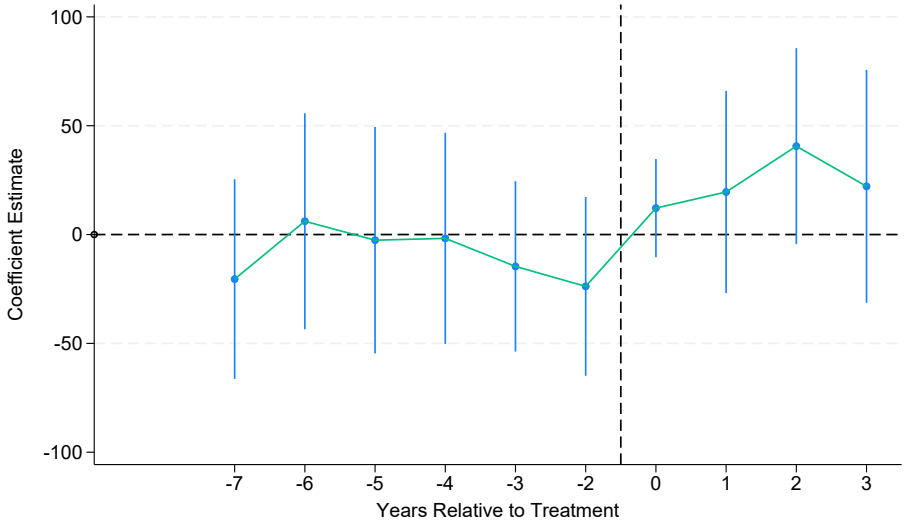


Figure 4: Burglary Event Study

Directionally, it appears assaults may slightly decrease (at a magnitude of around 20 per 100,000) in response to the Castle Law treatment, and burglary appears to increase slightly (at a magnitude of around 25 per 100,000). However, the blue confidence intervals tells us neither result is significant for any year.

5 Entropy Balancing: DiD

The following tables show the balance of the controls and dependent variable for each regression after using "ebalance" for entropy balancing. I only balanced the first moment (mean):

	Treat			Control		
	mean	variance	skewness	mean	variance	skewness
police_base	316.8	2054	.3709	316.8	2261	.2093
income_base	46602	4.31e+07	.8366	46622	2.76e+07	.478
exp_pubwe~se	1356	148126	2.113	1356	92546	1.684
assault_base	310.2	21193	.4178	310.2	13481	-.2402

Figure 5: Assault Entropy Balancing

	Treat			Control		
	mean	variance	skewness	mean	variance	skewness
police_base	316.8	2054	.3709	316.8	2256	.4162
income_base	46602	4.31e+07	.8366	46603	2.54e+07	.1804
exp_pubwe~se	1356	148126	2.113	1356	91855	2.176
burglary_b~e	816	42572	-.8644	816	99348	-.06693

Figure 6: Burglary Entropy Balancing

After balancing (in the 2005 year), the means of the treated and control match perfectly (except for a slight deviation in income_base).

I then run the same regression as before except with "aweight" to incorporate the entropy balanced weights:

assault	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
1.treated	.0307764	56.86449	0.00	1.000	-114.8094	114.871
1.after	-15.41552	5.326206	-2.89	0.006	-26.17201	-4.659027
treated#after						
1 1	4.527031	10.7786	0.42	0.677	-17.24079	26.29485
_cons	310.1737	40.42584	7.67	0.000	228.5321	391.8154

Figure 7: Assault DiD with Entropy balancing

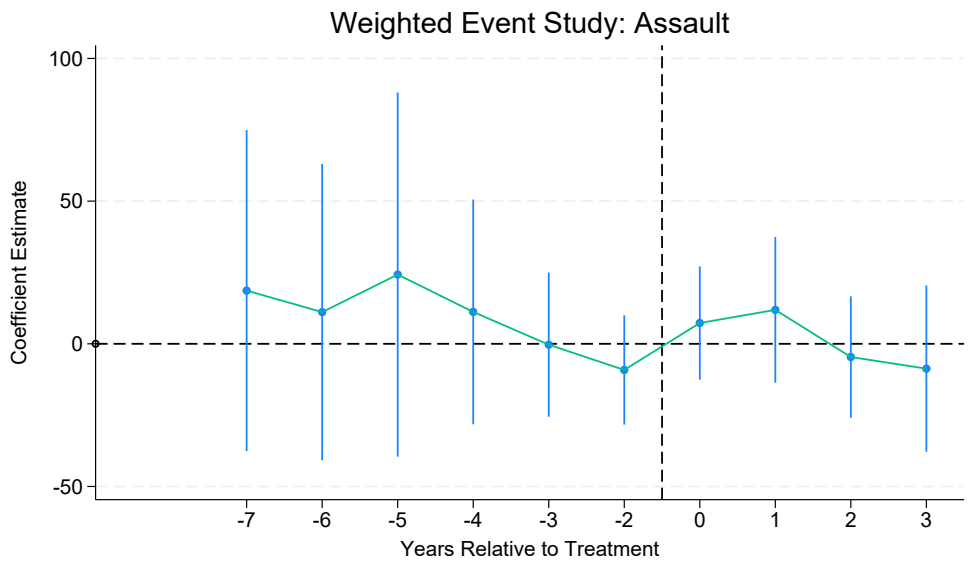
burglary	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
1.treated	-.0024929	145.2512	-0.00	1.000	-293.3432	293.3382
1.after	8.765148	40.94638	0.21	0.832	-73.92775	91.45805
treated#after						
1 1	13.50621	46.01563	0.29	0.771	-79.42424	106.4366
_cons	816.0145	133.7355	6.10	0.000	545.9301	1086.099

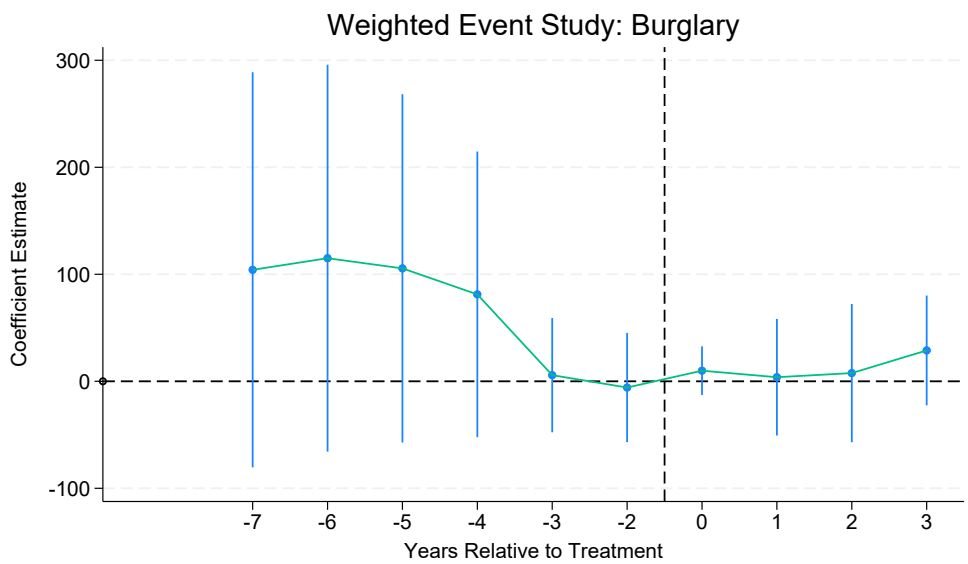
Figure 8: Burglary DiD with Entropy Balancing

In this case, both results are directionally positive - assaults and burglaries increase by a magnitude of 4.5 and 13.5 per 100,000 respectively - suggesting Castle Laws increase both assault and burglary. However, neither are significant.

6 Entropy Balancing: Event Study

Using the same weights as we did for DiD, we now do entropy balancing for our event studies. Using the same method outlined in the first event study (with the addition of "aweights") we get the following results:





Just as before in the event study, we get null results for all years. Strangely, the parallel trends in the pre-period look worse in the balanced event study plot for burglary than in the unbalanced plot. The magnitudes of the effect are near zero for both of these Event Studies.

7 Callaway & Sant'Anna DiD

Finally, because the data does contain treatments in other years I bring back the data of the states that are treated in other years and perform a staggered event study using the method outlined by Callaway & Sant'Anna (2020). To do this I design a "first_treatment_year" variable in my data and then run a csdid regression with clustered standard errors at the unit level and with "long2" so that the pre-treatment periods are long comparisons. Finally, I visualize the results using csdid_plot:

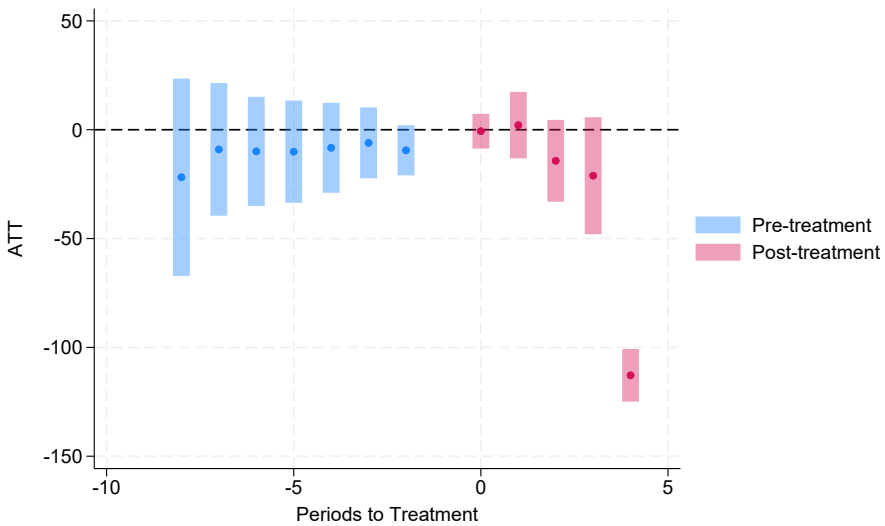


Figure 9: CSDID: Assault

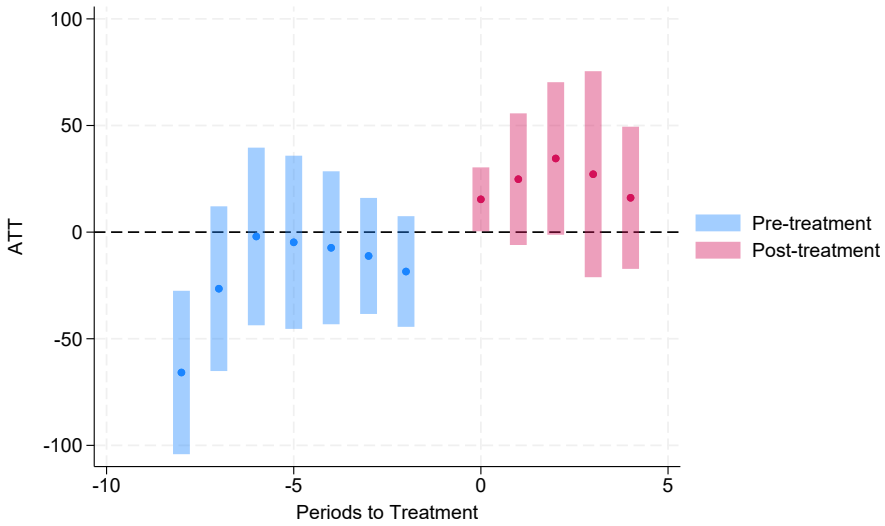


Figure 10: CSDID: Burglary

There are several things to note from these results. The first is that the burglary plot breaks the parallel trends assumption in the pre-treatment and has one significant post-treatment year. The one year of significance (in the first year of treatment), shows the passing of Castle Laws increases burglary incidents in that year by the magnitude of 15 per 100,000 people. The second thing to notice is that the assault model has a large and significant negative effect in the fifth year after treatment. To be specific, the model with assault as the dependent variable shows Castle Laws lower assaults by a magnitude of 112 per 100,000 people.